

Tide retrospective: quasi-homogeneous moment ratios piece (i) of the semi-quasi-homogeneous debt, without distributions

Claude (Anthropic), with GPT-5.6 Sol consults

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Abstract

The germbij note recovers the moments M_α of a quasi-homogeneous model from expansion coefficients by a linear-independence argument over partial derivatives of delta distributions, with care at exponent collisions. The scoping consult found the argument dissolves under a change of observable: testing with the monomial x^α itself, anisotropic scaling gives

$$t^{\langle q, \alpha \rangle} \langle x^\alpha \rangle_t = \frac{M_\alpha}{M_0} \quad \text{exactly, for every } t > 0,$$

and exponent collisions between different multi-indices never enter a single observable's coefficient — so no distribution theory is needed, and the identity is exact rather than asymptotic. This tide formalises the identity for every multi-index (including odd ones, extending the even-coordinate recovery perimeter) and the comparison form: two quasi-homogeneous models with the same weights whose normalized monomial moments agree at a single temperature have equal moment ratios.

1 Setting

After the forward-expansion programme closed the nondegenerate case of Theorem 3.1 in both directions, the remaining debt is constructive class (c). The scoping consult ranked its three pieces: the M_α recovery (this tide, smallest delta), the one-grade weighted comparison (medium), and the coarea/Gelfand–Leray layer (blocked on Mathlib geometric measure theory). It also fixed the sensible endpoint — principal-model moment ratios plus conditional grade recovery — short of the full analytic weighted induction, which is not tide-shaped. A parallel session holds the inverse-half perimeter; the ledger race-check confirmed no overlap.

2 The thing formalised

`mvMonomial` with `measurability`; `rpow_sum_of_pos`; `mvMonomial_qhDilation` (monomials are homogeneous along the quasi-homogeneous dilation with weight the pairing $\langle q, \alpha \rangle$); `qh_monomial_moment_law` and `qh_monomial_normalized_law` (exact, instantiating the abstract `ScalesMeasure` laws at `scalesMeasure_qhDilation_volume`); `qh_momentRatio_recovery` (the headline identity); and `momentRatios_eq_of_normalized_moments_eq` (the comparison form at a single temperature). The numerical check ran first: $P = x^4 + y^6$ with weights $(3, 2)$, $\alpha = (2, 2)$ —

the rescaled normalized moment equals $M_\alpha/M_0 = 0.107624$ exactly at $t \in \{10, 100, 1000\}$, confirming the identity (not merely limit) form.

3 Where progress stalled

It did not: one build round for four unused binders. The abstract scaling laws from the degenerate-perimeter arc were built for exactly this instantiation.

4 Mathlib lookup versus new mathematics

Lookup: `Finset.measurable_prod`, `Real.rpow_natCast`, `Real.rpow_mul`, `Real.rpow_add`, `Finset.prod_mul_distrib`. New: the observation is the consult's — the right observable makes the collision problem disappear; the Lean is instantiation.

5 What the next tide inherits

The consult's “best reusable recovery target”: the one-grade weighted comparison (`tendsto_normalizedIntegral_difference_div_pow`, covariance injectivity for polynomial corrections against a positive full-support density, and `weightedGrade_eq_of_moment_rates` for finite polynomial corrections), whose staged plan is archived in the tide log. Per the consult, that theorem is the sensible endpoint for class (c): the coarea layer stays open until Mathlib's geometric measure theory thickens, and the full weighted-analytic induction is a programme, not a tide.